Yang Zhang

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M.S. student in Computer Engineering at UC San Diego | LLM post-training, RL with verifiable rewards, and efficient serving

Education

University of California San Diego Dec. 2025 – Present
M.S. in Computer Engineering; GPA 3.94/4.00. Advisor: Ramamohan Paturi. La Jolla, CA

University of Electronic Science and Technology of China (UESTC) Sept. 2021 – Jun. 2025
B.Eng. in Communication Engineering; GPA 3.8/4.0 (top 10%). GRE 330; IELTS 8.0. Chengdu, China

Research Experience

UC San Diego, Laboratory for Emerging Intelligence 2026 – Present
Research Assistant (Advisor: Ramamohan Paturi) La Jolla, CA

RL for LLM reasoning

What rollout tokens buy in RLVR *Under review, ICLR 2027*

- Studied GRPO on competition mathematics with SmolLM3-3B (full-parameter, verl) and Qwen3.5-9B/35B-A3B (LoRA), scoring every model by majority-vote accuracy under a fixed budget of generated tokens per problem.
- Replacing one third of SmolLM3-3B's training problems with harder ones doubled response length (+101%) on the unchanged 2,048 problems at a similar training reward.
- Halving Qwen3.5-9B's training response limit from 4,096 to 2,048 tokens cut training compute by 28% and raised majority-vote accuracy at 16,384 tokens per problem from 36.1% to 43.5%.
- Showed that coverage and majority vote can rank a base model and its RL-trained version in opposite orders (57 vs. 39 problems with any correct sample; 5 vs. 18 correct majority answers).

QbQ: self-evolving curriculum for RL fine-tuning *arXiv 2026, under review at AAAI 2027; co-first author*

- Co-developed Question-begets-Question (QbQ): a teacher rewrites existing problems into diverse variants, and each round seeds new problems from those the current checkpoint can mostly solve ($8 \leq$ pass@16 ≤ 15).
- Trained Qwen2.5-Math-7B with GRPO on problem statements and final answers only, never on teacher reasoning traces; under the same 6,000-problem budget the curriculum lifted AIME pass@1 from 5.6% to 16.5%, past the 12.5–14.5% ceiling of static augmentation, with no saturation after 20 rounds.

Benchmarks and AI for science

CT Open: live, decontaminated clinical-trial forecasting *COLM 2026*

- Contributed to a live platform that runs four challenges a year and scores only predictions submitted before a trial's results became public; released a training set and two time-stamped test benchmarks (Winter 2025 and Summer 2025).
- Contributed to the automated decontamination pipeline (temporal filters, iterative LLM web search, expert review); a manual audit found no pre-cutoff results for 49 of 50 sampled trials.

DeepImagine: counterfactual imagination for clinical-trial prediction *Under review, ICLR 2027*

- Predicts a target trial's outcome from a historical trial by changing one differing factor at a time (enrollment, study arm, outcome measure); with seven anchors, lifts GPT-5 from 72.4 (RAG) to 79.0 macro-F1 on 701 CT Open questions.
- Contributed to per-factor Gemma-3-12B specialists trained on natural counterfactuals from 14,000 trials with GPT-5 reasoning traces; the trained system reaches 78.7 macro-F1, above all ten conventional and clinical predictors.

Interpretability

Latent Assembly: executable operations in language models *Under review, ICLR 2027*

- Contributed to a causal framework that recovers typed program states and state-dependent operations from unlabeled activations, validated on compiled transformers and on SELECT and LOAD operations in Qwen, Gemma and Llama.
- LOAD, ADD and COMPARE operations discovered independently on separate tasks in Qwen2.5-14B link into a chain that executes a held-out program at 84% accuracy vs. 49% zero-shot, with no target-task fitting.

Efficient LLM serving (collaboration with Harvard University) 2026
LoopKV: loop-aware KV cache compression for looped transformers arXiv 2026; co-first author

- Characterized Ouro's per-loop KV cache: quantization sensitivity varies per (loop, layer) cell and adds up to the output KL, later loops tolerate sharing the final loop's KV, and read errors compound across loops (4.7× less damage at equal bytes when low-attention values are read at 2 bits rather than skipped).
- Co-designed an offline profiler and knapsack planner that assign INT4, INT2 or KV sharing to each cell, attention-aware V precision at decode time, and one fused vLLM kernel that executes every cell type in a single launch (host launches per decode step from 504 to 17 with CUDA graphs).
- On Ouro-1.4B (one RTX PRO 6000), LoopKV-M stores 35% less KV than uniform INT4 (KIVI-4) and sustains 1.7–2.2× its goodput under a 16K latency SLO; 1.13–4.44× BF16 decode throughput on three looped models with no suite average significantly below BF16.

Institute of Computing Technology, Chinese Academy of Sciences Jul. 2024 – Aug. 2024
Research Assistant, State Key Laboratory of Processors Beijing, China

- Developed and optimized image signal processing algorithms (noise reduction, sharpening, color correction) in Python, MATLAB and C for embedded systems built around CMOS image sensors.

Institute of Computing Technology, Chinese Academy of Sciences Aug. 2023 – Nov. 2023
Research Assistant, Center for Advanced Computer Systems Beijing, China

- Improved the Maximum Information Coefficient (MIC) algorithm for identifying and visualizing correlations across datasets, and optimized the data-processing workflow for accuracy and efficiency.

Publications

* equal contribution

Under review and preprints

[arXiv 2026] LoopKV: Loop-Aware KV Cache Compression for Serving Looped Transformers
Tianle Zhou*, **Yang Zhang***, Tianyu Wu*, Zhuohan Wang*, William Nixon, Juncheng Yang.
arXiv preprint arXiv:2611.04137, 2026. [arXiv]

[Under review] What Do Rollout Tokens Buy? Hidden Costs and Budgeted Accuracy in RLVR
Yang Zhang, et al. (co-authors withheld during double-blind review).
Under review at the International Conference on Learning Representations (ICLR) 2027.

[Under review] Latent Assembly: Causal Discovery and Cross-Task Linking of Executable Operations in Language Models
Yang Zhang, et al. (co-authors withheld during double-blind review).
Under review at ICLR 2027.

[Under review] DeepImagine: Clinical Trial Outcome Prediction via Stepwise Local Counterfactual Imaginations
Yang Zhang, et al. (co-authors withheld during double-blind review).
Under review at ICLR 2027. arXiv:2604.23054. [arXiv] [Code]

[arXiv 2026] Question Begets Question: Self-Evolving Curriculum for Reinforcement Fine-Tuning on Competition Mathematics
Longtian Bao*, Jianyou Wang*, **Yang Zhang***, Youze Zheng*, Ramamohan Paturi.
arXiv preprint arXiv:2608.01522, 2026; under review at AAAI 2027. [arXiv]

Conference papers

[COLM 2026] CT Open: An Open-Access, Uncontaminated, Live Platform for the Open Challenge of Clinical Trial Outcome Prediction
Jianyou Wang*, Youze Zheng*, Longtian Bao*, Hanyuan Zhang*, Qirui Zheng, Yuhan Chen, **Yang Zhang**, Matthew Feng, Maxim Khan, Aditya K. Sehgal, Christopher D. Rosin, Ramamohan Paturi, Umber Dube, Leon Bergen.
Conference on Language Modeling (COLM), 2026. [arXiv] [Benchmark] [Code]

[ICAACE 2025] Improved RT-DETR Based on MobileNetV4 for Vehicle Detection
Yang Zhang.
2025 8th International Conference on Advanced Algorithms and Control Engineering (ICAACE), IEEE, 2025, pp. 901–906. [IEEE Xplore]

Skills

- **Post-training:** GRPO/RLVR (verl), SFT, LoRA, synthetic data and self-evolving curricula, pass@k and majority-vote evaluation under token budgets, benchmark construction and decontamination, LLM-as-judge.
- **Serving and systems:** vLLM, KV-cache quantization and sharing, fused attention kernels, CUDA graphs, throughput/goodput measurement under latency SLOs.
- **Tools:** PyTorch, Hugging Face Transformers, vLLM, NumPy, pandas, scikit-learn, Git | **Languages:** Python, C, MATLAB, Bash, LaTeX.
- **Computer vision:** real-time object detection (RT-DETR with a MobileNetV4 backbone: 97.3 mAP50 at 11M parameters and 27 GFLOPs, 65% fewer parameters and 75% less compute than RT-DETR-L; the Medium variant tops 23 compared detectors at 98.4 mAP50); image signal processing.

Honors and Awards

1st Place, UCSD x Medeloop Health Research Competition, 2026 | UESTC Merit Scholarship for Exceptional Student (top 10%), 2024 | UESTC Merit Scholarship (top 30%), 2022 and 2023 | Hainan Free Trade Port Merit Scholarship (top 10%), 2022

Service

Reviewer, ICLR 2027 | Reviewer, AAAI 2027 | Student Volunteer (session monitor), ACL 2026, San Diego, CA